

Task1_Python_File_Automation

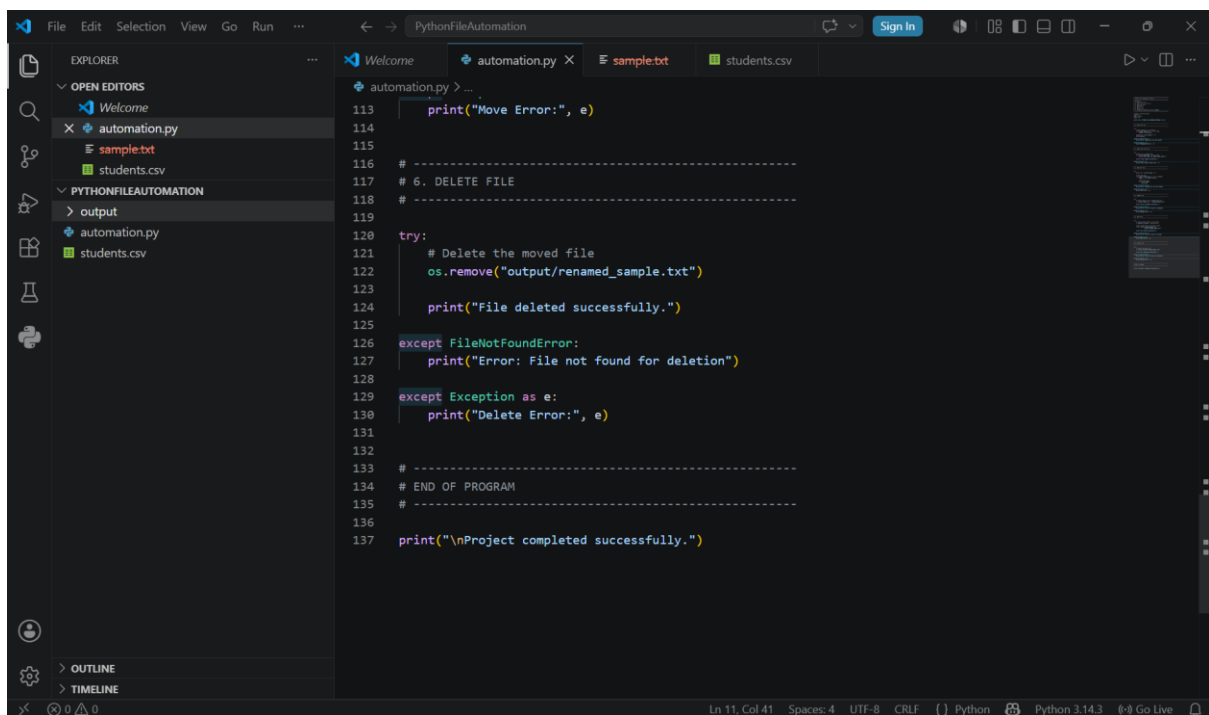
Objective

The objective of this project is to understand Python file handling, automation logic, and exception handling. This project demonstrates how to perform file operations such as reading, writing, renaming, moving, and deleting files using Python. It also helps in learning how to manage errors effectively using try-except blocks.

Technologies Used

- Python
- Flask
- Pandas
- VS Code
- GitHub
- Render

Code Screenshots



The screenshot shows a Visual Studio Code editor window with the file explorer on the left and a Python script open in the main editor. The script is titled 'automation.py' and contains the following code:

```
113 print("Move Error:", e)
114
115
116 # -----
117 # 6. DELETE FILE
118 # -----
119
120 try:
121     # Delete the moved file
122     os.remove("output/renamed_sample.txt")
123
124     print("File deleted successfully.")
125
126 except FileNotFoundError:
127     print("Error: File not found for deletion")
128
129 except Exception as e:
130     print("Delete Error:", e)
131
132
133 # -----
134 # END OF PROGRAM
135 # -----
136
137 print("\nProject completed successfully.")
```

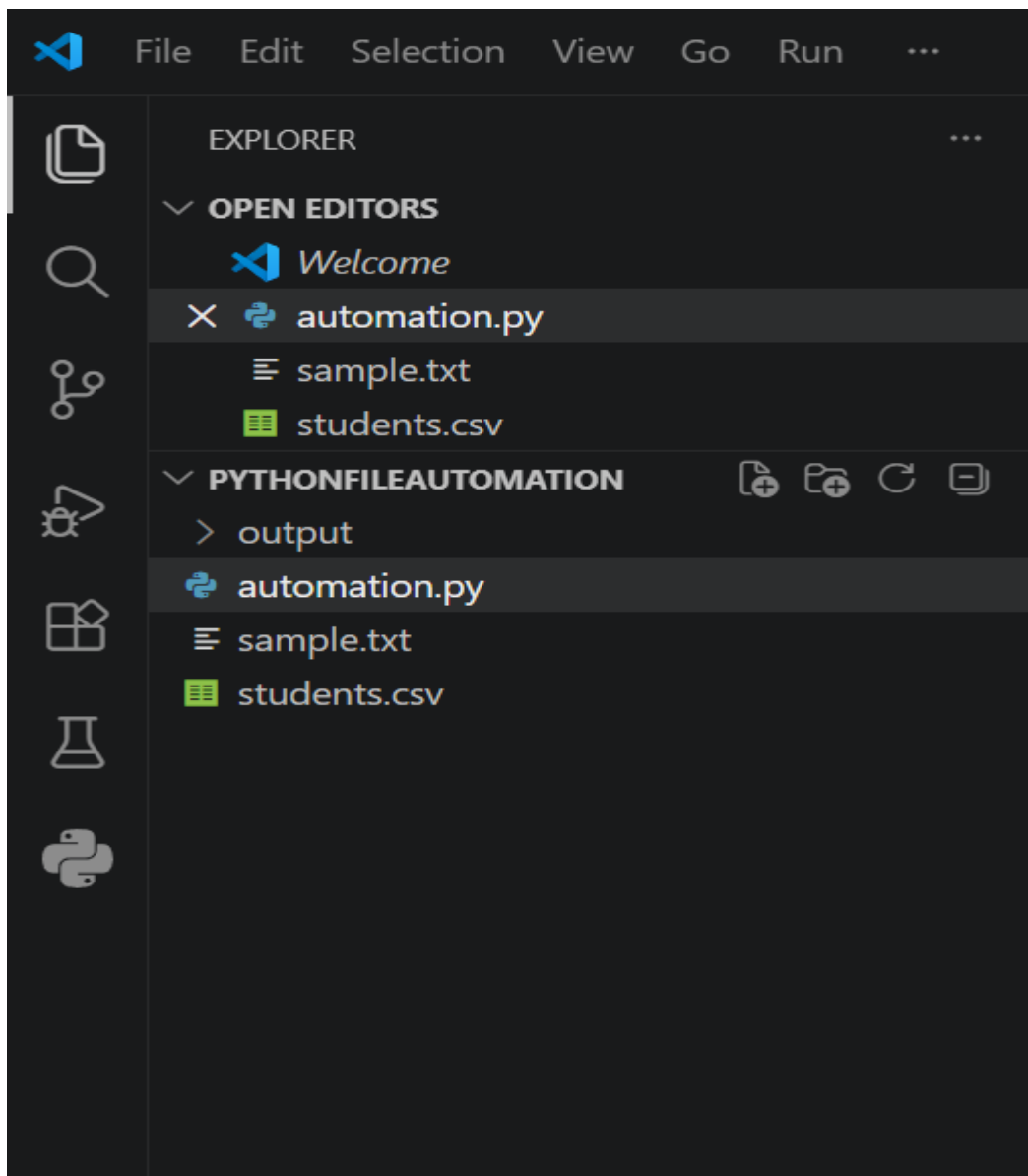
The file explorer on the left shows the project structure with folders 'PYTHONFILEAUTOMATION' and 'output', and files 'automation.py', 'sample.txt', and 'students.csv'. The status bar at the bottom indicates the current line and column (Ln 11, Col 41) and the file encoding (UTF-8).

```
85     print("\nFile renamed successfully.")
86
87 except FileNotFoundError:
88     print("Error: File not found for renaming")
89
90 except Exception as e:
91     print("Rename Error:", e)
92
93
94 # -----
95 # 5. MOVE FILE
96 # -----
97
98 try:
99     # Create output folder if not exists
100     os.makedirs("output", exist_ok=True)
101
102     # Move renamed file into output folder
103     shutil.move("renamed_sample.txt",
104               "output/renamed_sample.txt")
105
106     print("File moved successfully.")
107
108 except FileNotFoundError:
109     print("Error: File not found for moving")
110
111 except Exception as e:
112     print("Move Error:", e)
113
114
115 # -----
116 # 6. DELETE FILE
117
```

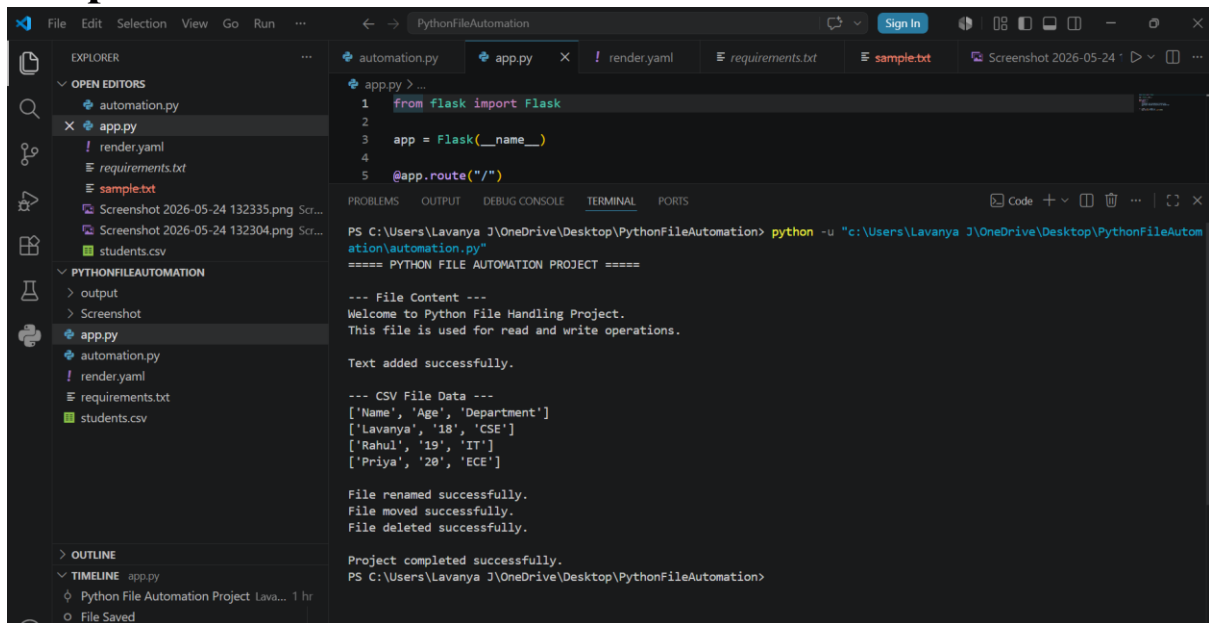
```
57 # 3. READ CSV FILE
58 # -----
59
60 try:
61     print("\n--- CSV File Data ---")
62
63     # Open CSV file
64     with open("students.csv", "r") as csvfile:
65         reader = csv.reader(csvfile)
66
67         # Read each row
68         for row in reader:
69             print(row)
70
71 except FileNotFoundError:
72     print("Error: students.csv file not found")
73
74 except Exception as e:
75     print("CSV Error:", e)
76
77
78 # -----
79 # 4. RENAME FILE
80 # -----
81
82 try:
83     # Rename sample.txt to renamed_sample.txt
84     os.rename("sample.txt", "renamed_sample.txt")
85
86     print("\nFile renamed successfully.")
87
88 except FileNotFoundError:
89     print("Error: File not found for renaming")
90
```

```
31 print("\n--- File Content ---")
32 print(content)
33
34 except FileNotFoundError:
35     print("Error: sample.txt file not found")
36
37 except Exception as e:
38     print("Unexpected Error:", e)
39
40 # -----
41 # 2. WRITE INTO TEXT FILE
42 # -----
43
44 try:
45     # Open file in append mode
46     with open("sample.txt", "a") as file:
47         file.write("\nNew line added using Python.")
48
49     print("\nText added successfully.")
50
51 except Exception as e:
52     print("Error while writing file:", e)
53
54 # -----
55 # 3. READ CSV FILE
56 # -----
57
58 try:
59     print("\n--- CSV File Data ---")
60
61     # Open CSV file
```

```
1 # =====
2 # Python File Automation Project
3 # =====
4 # Features:
5 # 1. Read text file
6 # 2. Write into a text file
7 # 3. Read CSV file
8 # 4. Rename file
9 # 5. Move file
10 # 6. Delete file
11 # 7. Exception handling using try-except
12 # =====
13
14 # Import required modules
15 import os
16 import shutil
17 import csv
18
19 print("==== PYTHON FILE AUTOMATION PROJECT =====")
20
21 # -----
22 # 1. READ TEXT FILE
23 # -----
24
25 try:
26     # Open sample.txt in read mode
27     with open("sample.txt", "r") as file:
28         content = file.read()
29
30     print("\n--- File Content ---")
31     print(content)
32
33 except FileNotFoundError:
```



Output Screenshots



The screenshot shows the Visual Studio Code interface with the `app.py` file open in the editor. The file contains the following code:

```
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route("/")
```

The terminal output shows the execution of the script:

```
PS C:\Users\Lavanya J\OneDrive\Desktop\PythonFileAutomation> python -u "c:\Users\Lavanya J\OneDrive\Desktop\PythonFileAutomation\automation.py"
===== PYTHON FILE AUTOMATION PROJECT =====

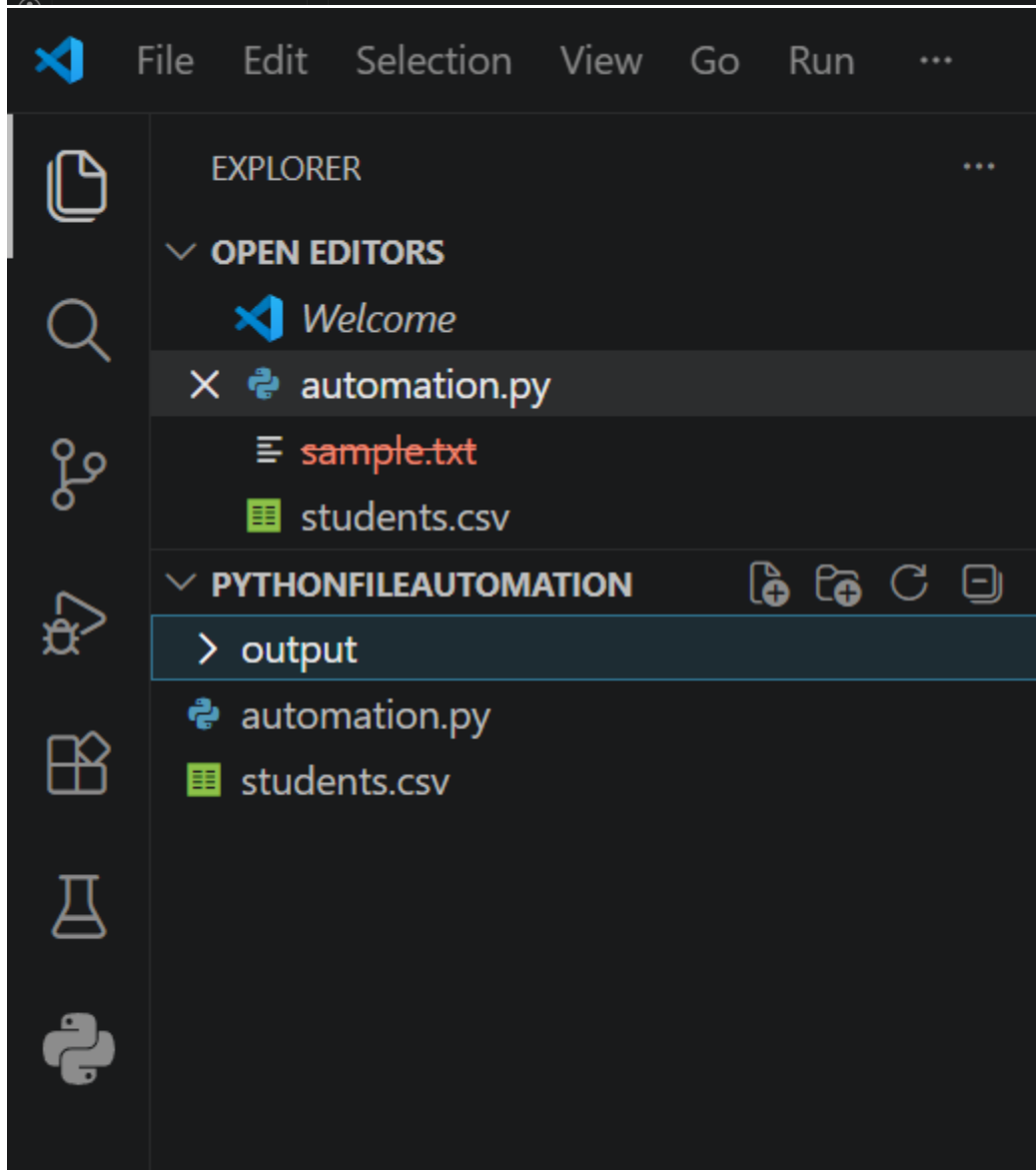
--- File Content ---
Welcome to Python File Handling Project.
This file is used for read and write operations.

Text added successfully.

--- CSV File Data ---
['Name', 'Age', 'Department']
['Lavanya', '18', 'CSE']
['Rahul', '19', 'IT']
['Priya', '20', 'ECE']

File renamed successfully.
File moved successfully.
File deleted successfully.

Project completed successfully.
PS C:\Users\Lavanya J\OneDrive\Desktop\PythonFileAutomation>
```



GitHub Repository Link

<https://github.com/jlavanya132008-byte/python-file-automation>

Hosted Link

<https://python-file-automation-1.onrender.com>

Conclusion

This project helped me understand the basics of Python file handling and automation.